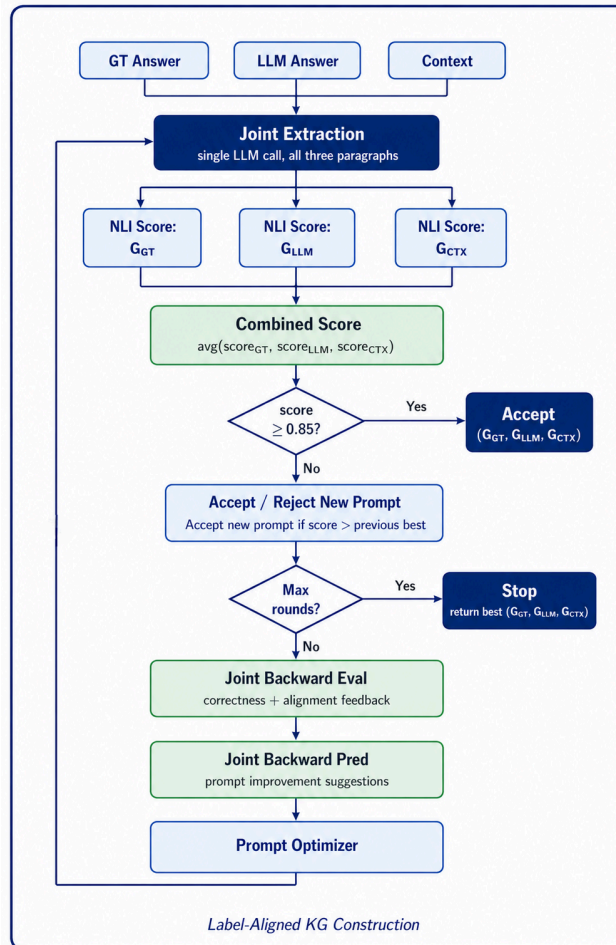


KGC

Overview



Extracting multiple label signed accurate knowledge graphs. Follows the KRPO style refinement.

Why

- Joint extraction prompt encourages shared labels across graphs, making graphs suitable for downstream cross comparisons.
- KRPO style refinement, detects hallucinations/contradictions and helps to remove them, making accurate graphs.

Goals:

- Improved factual accuracy of extracted triplets.
- Keep entity and relation labels aligned across all graphs.

Step 1 — Joint Extraction

With the prompt engineered extraction prompt, a LLM is prompted to to extract 3 KGs accurately and sharing common entity/relation naming across graphs

Step 2 — Per-Graph NLI Scoring

Each extracted graph is scored against its own source paragraph. Every triplet is converted to a natural-language sentence and evaluated by an NLI model.

How it works:

- For each triplet (subject, relation, object), a triplet-to-text prompt generates the triplet sentence.
- The NLI model receives the source paragraph and the triplet sentence.
- It classifies the triplet as Entailment, Neutral, or Contradiction.
- Each label maps to a numeric reward.

Label	Reward	Meaning
Entailment	+1.0	Triplet logically follows from the source paragraph
Neutral	0.0	Neither supported nor contradicted
Contradiction	-0.5	Triplet conflicts with the source paragraph

Per-graph score formula:

```
score_k = (entailments - 0.5 * contradictions) / total_triplets
```

Step 3 — Combined Score

The per-graph scores are averaged into a single combined score. If this score already exceeds the threshold, the loop exits early and returns the current best graphs.

```
combined = avg(score_GT, score_LLM, score_CTX)
```

- If combined >= 0.85 -> accept current graphs and stop.
- Otherwise continue to the feedback steps.

Step 4 — Joint Backward Feedback

Backward Eval identifies recurring failure patterns across all graphs. A LLM prompt

- Input: extracted triplets, per-triplet NLI labels, combined score.
- Performs pattern-level analysis, not a raw triplet dump.
- Hallucinations: inferred from triplets repeatedly marked neutral or contradiction.
- Missing facts: inferred when source content is absent from the extracted graphs.

- Label mismatch: inferred when semantically equivalent entities or relations use inconsistent names across graphs.

Backward Pred proposes concrete prompt improvements.

- Input: current system prompt, all source texts, extracted graphs, Backward Eval feedback.
- Output: actionable guidance on instruction clarity, coverage, and alignment constraints.

Step 5 — Prompt Optimizer

Rewrites the extraction prompt using the feedback from Step 4.

- Takes the current prompt as a variable alongside contextual feedback examples.
- Rewrites instruction wording while preserving strict output format constraints.
- Produces a candidate new extraction prompt — not yet the active prompt.

Accept / Reject new extraction prompt

The candidate prompt is tested immediately before it replaces the active prompt.

- Run a trial joint extraction using the candidate prompt.
- Score the trial graphs using the same NLI pipeline as Step 2.
- If `trial_combined > best_combined_score`: adopt the new prompt and update the best graphs.
- Otherwise: discard the candidate and keep the previous prompt.

This gate prevents a bad rewrite from overwriting a better prompt.

Step 7 — Repeat

The loop continues with either the updated or the retained prompt until the early-stop threshold is hit or max rounds are reached.

- Each iteration uses the accepted prompt (if improved) or the previous prompt (if rejected).
- Final output is the best-scoring graphs found during optimization, not necessarily the last-round output.

Extra than in base KRPO

- Incorporates label alignment across graphs, by joint extraction, separate NLI scores, combines score
- Other than that same KRPO components

Evaluation

Jaccard similarity between labels across generated graphs
Accuracy with F1

Method	Strict F1	Label Jaccard	H
KRPO (single-graph)	0.707	0.426	0.530
Ours	0.611	0.572	0.591
KEA Baseline	0.518	0.649	0.576

KRPO single extraction for independent graphs- best accuracy, low label alignment
KEA - joint single prompt extraction - best alignment, low accuracy(often hallucinations)
Ours - Best balance between accuracy and label alignment (high harmonic mean)